

Worksheet 3

Student Name: Navjot Kaur

Branch: MCA

Semester: 3rd

Subject Name: Internet of Things

UID: 25MCA20078

Section/Group: 25MCA-1(A)

Date of Performance: 31-08-2026

Subject Code: 25CAH-708

1. Aim/Overview of the practical:

Interface an Arduino Uno with Ultrasonic Sensor to calculate the distance and print on a Serial monitor and on LCD. Follow the mentioned cases after the calculation of distance.

- If the distance is > 100 inches green light is in ON condition.
- If the distance is < 100 inches and the distance > 60 yellow light is in ON condition and the green light remains Off
- If the distance is < 60 inches turn on the buzzer and the lights will remain OFF.

2. Apparatus (For applied/experimental sciences/materials based labs):

Hardware Requirements

- Arduino Uno
- Ultrasonic Sensor (HC-SR04)
- 16×2 LCD
- Green LED
- Yellow LED
- Buzzer
- Resistors (220Ω for LEDs)
- Breadboard and jumper wires

Software requirements

- Arduino IDE
- Tinkercad Circuits

3. Circuit Diagram (TinkerCad):

Code:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
```

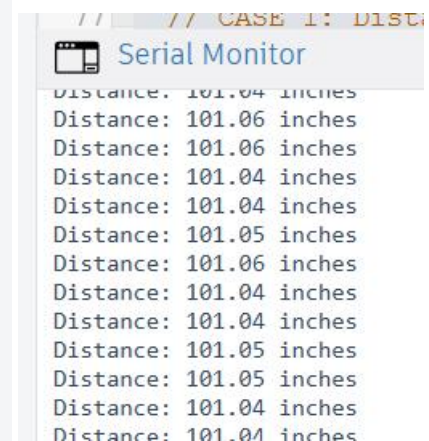
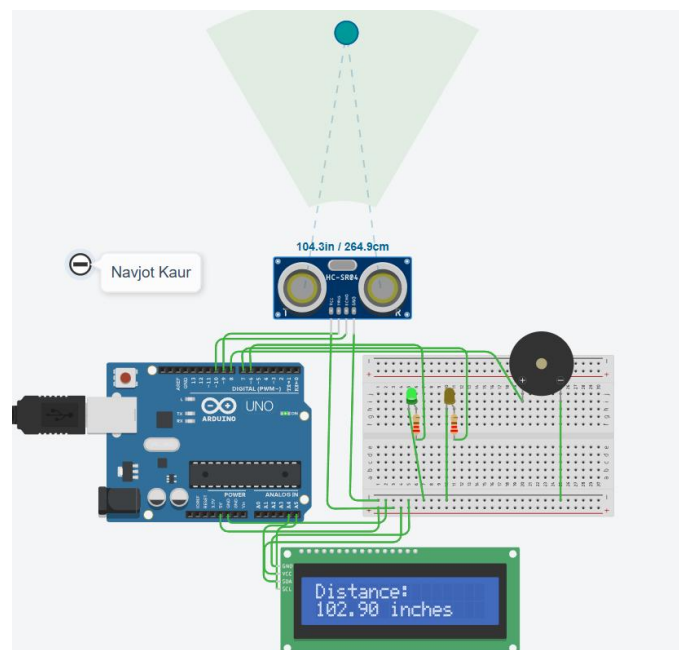
```
// I2C LCD address
LiquidCrystal_I2C lcd(0x27, 16, 2);
// Pin definitions
const int trigPin = 9;
const int echoPin = 10;
const int greenLED = 6;
const int yellowLED = 7;
const int buzzer = 8;
long duration;
float distanceCM;
float distanceInch;
void setup() {
  Serial.begin(9600);
  // Ultrasonic sensor pins
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  // Output pins
  pinMode(greenLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(buzzer, OUTPUT);
  // LCD setup
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Distance Meter");
  delay(2000);
  lcd.clear();
}
void loop() {
  // Send ultrasonic signal
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  // Calculate duration
  duration = pulseIn(echoPin, HIGH);
  // Calculate distance in cm
  distanceCM = duration * 0.034 / 2;
  // Convert cm to inches
  distanceInch = distanceCM / 2.54;
  // Print on Serial Monitor
  Serial.print("Distance: ");
  Serial.print(distanceInch);
  Serial.println(" inches");
  // Display on LCD
```

```

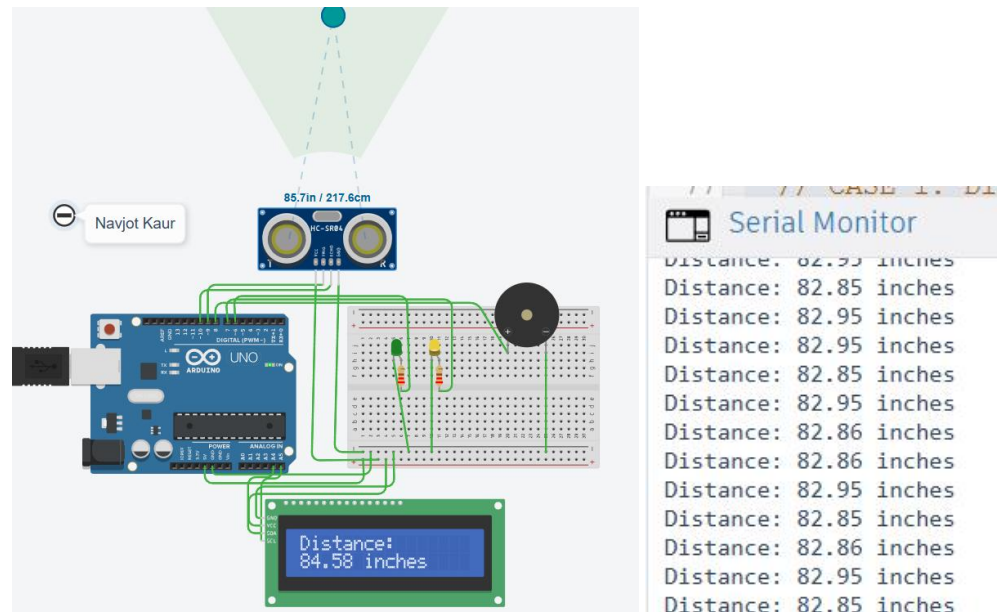
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Distance:");
lcd.setCursor(0, 1);
lcd.print(distanceInch);
lcd.print(" inches");
// CASE 1: Distance greater than or equal to 100 inches
if (distanceInch >= 100) {
    digitalWrite(greenLED, HIGH);
    digitalWrite(yellowLED, LOW);
    noTone(buzzer);
}
// CASE 2: Distance between 60 and 100 inches
else if (distanceInch >= 60 && distanceInch < 100) {
    digitalWrite(greenLED, LOW);
    digitalWrite(yellowLED, HIGH);
    noTone(buzzer);
}
// CASE 3: Distance less than 60 inches
else {
    digitalWrite(greenLED, LOW);
    digitalWrite(yellowLED, LOW);
    tone(buzzer, 1000);
}
delay(500);
}

```

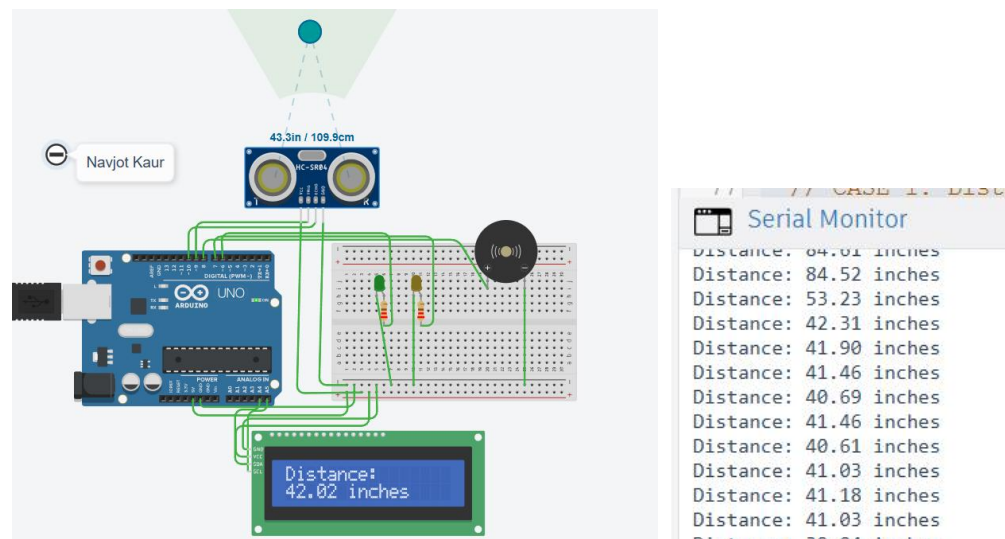
Case 1 Output:



Case 2 Output:



Case 3 Output:



5. Learning outcomes (What I have learnt):

1. Learned to interface an ultrasonic sensor with Arduino.
2. Learned to measure distance using the sensor.
3. Learned to display distance on LCD and Serial Monitor.
4. Controlled LEDs and a buzzer based on distance.